1.54 Equations of Lines in 3D - Vector Equations

Worked Solutions

1 a)

$$\mathbf{r} = \begin{pmatrix} 2 \\ -7 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 9 \\ -4 \end{pmatrix}.$$

1 b)

$$\mathbf{r} = \begin{pmatrix} 3 \\ 8 \\ -3 \end{pmatrix} + \mu \begin{pmatrix} -4 \\ -2 \\ 7 \end{pmatrix}.$$

2 a)

$$\mathbf{d} = \begin{pmatrix} 4 \\ -1 \\ 3 \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 4 \\ -1 \\ 3 \end{pmatrix}.$$

$$\mathbf{r} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ -1 \\ 3 \end{pmatrix}$$

or $\mathbf{r} = \begin{pmatrix} 4 \\ -1 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ -1 \\ 3 \end{pmatrix}$ or just $\mathbf{r} = \lambda \begin{pmatrix} 4 \\ -1 \\ 3 \end{pmatrix}$.

2 b)

$$\mathbf{d} = \begin{pmatrix} 3 \\ 5 \\ 7 \end{pmatrix} - \begin{pmatrix} 1 \\ 4 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ 6 \end{pmatrix}.$$

$$\mathbf{r} = \begin{pmatrix} 1 \\ 4 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ 6 \end{pmatrix}$$

or $\mathbf{r} = \begin{pmatrix} 3 \\ 5 \\ 7 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ 6 \end{pmatrix}$.

2 c)

$$\mathbf{d} = \begin{pmatrix} 3 \\ -1 \\ 6 \end{pmatrix} - \begin{pmatrix} 2 \\ -4 \\ 5 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix}.$$

$$\mathbf{r} = \begin{pmatrix} 2 \\ -4 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix}$$

or $\mathbf{r} = \begin{pmatrix} 3 \\ -1 \\ 6 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix}$.

2 d)

$$\mathbf{d} = \begin{pmatrix} 9 \\ -3 \\ -8 \end{pmatrix} - \begin{pmatrix} 9 \\ -8 \\ -2 \end{pmatrix} = \begin{pmatrix} 0 \\ 5 \\ -6 \end{pmatrix}.$$

$$\mathbf{r} = \begin{pmatrix} 9 \\ -8 \\ -2 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ 5 \\ -6 \end{pmatrix}$$

or $\mathbf{r} = \begin{pmatrix} 9 \\ -3 \\ -8 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ 5 \\ -6 \end{pmatrix}$.

2 e)

$$\mathbf{d} = \begin{pmatrix} 1 \\ -4 \\ 8 \end{pmatrix} - \begin{pmatrix} 6 \\ -4 \\ 13 \end{pmatrix} = \begin{pmatrix} -5 \\ 0 \\ -5 \end{pmatrix}.$$

$$\mathbf{r} = \begin{pmatrix} 6 \\ -4 \\ 13 \end{pmatrix} + \lambda \begin{pmatrix} -5 \\ 0 \\ -5 \end{pmatrix}$$

or $\mathbf{r} = \begin{pmatrix} 1 \\ -4 \\ 8 \end{pmatrix} + \lambda \begin{pmatrix} -5 \\ 0 \\ -5 \end{pmatrix}$, or divide the direction vector through by -5 to get $\mathbf{r} = \begin{pmatrix} 6 \\ -4 \\ 13 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$
or $\mathbf{r} = \begin{pmatrix} 1 \\ -4 \\ 8 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$.

2 f)

$$\mathbf{d} = \begin{pmatrix} -9 \\ -4 \\ 3 \end{pmatrix} - \begin{pmatrix} -1 \\ 10 \\ 3 \end{pmatrix} = \begin{pmatrix} -8 \\ -14 \\ 0 \end{pmatrix}.$$

$$\mathbf{r} = \begin{pmatrix} -1 \\ 10 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} -8 \\ -14 \\ 0 \end{pmatrix}$$

or $\mathbf{r} = \begin{pmatrix} -9 \\ -4 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} -8 \\ -14 \\ 0 \end{pmatrix}$, or divide the direction vector through by -2 to get $\mathbf{r} = \begin{pmatrix} -1 \\ 10 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ 7 \\ 0 \end{pmatrix}$
or $\mathbf{r} = \begin{pmatrix} -9 \\ -4 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ 7 \\ 0 \end{pmatrix}$.

3)

$\lambda = 2$:

$$\overrightarrow{OA} = \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} + 2\begin{pmatrix} -2 \\ 3 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 5 \\ 4 \end{pmatrix}.$$

$\lambda = -2$:

$$\overrightarrow{OB} = \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} - 2\begin{pmatrix} -2 \\ 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 7 \\ -7 \\ 0 \end{pmatrix}.$$

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} 7 \\ -7 \\ 0 \end{pmatrix} - \begin{pmatrix} -1 \\ 5 \\ 4 \end{pmatrix} = \begin{pmatrix} 8 \\ -12 \\ -4 \end{pmatrix}.$$

$$|\overrightarrow{AB}| = \sqrt{8^2 + (-12)^2 + (-4)^2} = \sqrt{224} = 4\sqrt{14} \text{ units.}$$

4)

$$6 = 5 - 3\lambda \Rightarrow \lambda = -\frac{1}{3}, \qquad -4 = -2 + 6\lambda \Rightarrow \lambda = -\frac{1}{3}, \qquad -12 = -3 + 27\lambda \Rightarrow \lambda = -\frac{1}{3}.$$

Consistent values of λ, therefore $(6, -4, -12)$ lies on the line.

5)

$$-12 = -2 + 4\lambda \Rightarrow \lambda = -\frac{5}{2}, \qquad \frac{31}{2} = 13 - \lambda \Rightarrow \lambda = -\frac{5}{2}, \qquad -\frac{27}{2} = -6 + 7\lambda \Rightarrow \lambda = -\frac{15}{4}.$$

Inconsistent values of λ, therefore $(-12, \frac{31}{2}, -\frac{47}{2})$ does not lie on the line.

6)

$$-2 = 7.5 + \lambda \Rightarrow \lambda = -\frac{19}{2}, \qquad 16 = -3 - 2\lambda \Rightarrow \lambda = -\frac{19}{2}, \qquad -40 = 2.5 + 5\lambda \Rightarrow \lambda = -\frac{17}{2}.$$

Inconsistent values of λ, therefore $(-2, 16, -40)$ does not lie on the line.

7)

$$5 = 2 + \lambda \Rightarrow \lambda = 3, \qquad -10 = -4 - 2\lambda \Rightarrow \lambda = 3, \qquad 2 = k + (3) \times 3 \Rightarrow k = -7.$$

8)

$$-3 = 4 + k\lambda \quad (1), \qquad -83 = -3 + 4\lambda \Rightarrow \lambda = -20, \qquad -98 = 2 + 5\lambda \Rightarrow \lambda = -20.$$

Substitute $\lambda = -20$ into (1): $-3 = 4 + k(-20) \Rightarrow k = \frac{7}{20}$.

9)

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} -5 \\ -9 \\ -1 \end{pmatrix} - \begin{pmatrix} 1 \\ -1 \\ -5 \end{pmatrix} = \begin{pmatrix} -6 \\ -8 \\ 4 \end{pmatrix}.$$

So the line passing through A and B has equation $\mathbf{r} = \begin{pmatrix} 1 \\ -1 \\ -5 \end{pmatrix} + \lambda \begin{pmatrix} -6 \\ -8 \\ 4 \end{pmatrix}$. Check if C lies on the line:

$$\begin{pmatrix} 16 \\ 19 \\ -15 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ -5 \end{pmatrix} + \lambda \begin{pmatrix} -6 \\ -8 \\ 4 \end{pmatrix}.$$

$$16 = 1 - 6\lambda \Rightarrow \lambda = -\frac{5}{2}, \quad 19 = -1 - 8\lambda \Rightarrow \lambda = -\frac{5}{2}, \quad -15 = -5 + 4\lambda \Rightarrow \lambda = -\frac{5}{2}.$$

Consistent values of λ, therefore A, B and C are collinear (they lie on the same line).

10)

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} -8 \\ 18 \\ -10 \end{pmatrix} - \begin{pmatrix} 40 \\ -46 \\ -18 \end{pmatrix} = \begin{pmatrix} -48 \\ 64 \\ 8 \end{pmatrix}.$$

So the line passing through A and B has equation $\mathbf{r} = \begin{pmatrix} 40 \\ -46 \\ -18 \end{pmatrix} + \lambda \begin{pmatrix} -48 \\ 64 \\ 8 \end{pmatrix}$. Check if C lies on the line:

$$\begin{pmatrix} 4 \\ 2 \\ -11 \end{pmatrix} = \begin{pmatrix} 40 \\ -46 \\ -18 \end{pmatrix} + \lambda \begin{pmatrix} -48 \\ 64 \\ 8 \end{pmatrix}.$$

$$4 = 40 - 48\lambda \Rightarrow \lambda = \frac{3}{4}, \quad 2 = -46 + 64\lambda \Rightarrow \lambda = \frac{3}{4}, \quad -11 = -18 + 8\lambda \Rightarrow \lambda = \frac{7}{8}.$$

Inconsistent values of λ, therefore A, B and C are not collinear (they do not lie on the same line).

11)

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} -1 \\ 11 \\ 27 \end{pmatrix} - \begin{pmatrix} 7 \\ p \\ p \end{pmatrix} = \begin{pmatrix} -8 \\ 11 - p \\ 27 - p \end{pmatrix}.$$

So the line passing through A and B has equation $\mathbf{r} = \begin{pmatrix} 7 \\ p \\ p \end{pmatrix} + \lambda \begin{pmatrix} -8 \\ 11 - p \\ 27 - p \end{pmatrix}$. C lies on the line:

$$\begin{pmatrix} 17 \\ -16 \\ q \end{pmatrix} = \begin{pmatrix} 7 \\ p \\ p \end{pmatrix} + \lambda \begin{pmatrix} -8 \\ 11 - p \\ 27 - p \end{pmatrix}.$$

$$17 = 7 - 8\lambda \Rightarrow \lambda = -\frac{5}{4}.$$

$$-16 = p - \frac{5}{4}(11 - p) \Rightarrow -64 = 4p - 55 + 5p \Rightarrow 9p = -9 \Rightarrow p = -1.$$

$$q = -1 - \frac{5}{4}(27 - (-1)) = -36.$$

Therefore $p = -1$ and $q = -36$.

12)

$$-2 = k(0.5) \Rightarrow k = -4, \quad 8 = k(-2) \Rightarrow k = -4, \quad 13.5 = k(-3.375) \Rightarrow k = -4.$$

$$\therefore \begin{pmatrix} -2 \\ 8 \\ 13.5 \end{pmatrix} = -4 \begin{pmatrix} 0.5 \\ -2 \\ -3.375 \end{pmatrix}.$$

Check if $(7, 8, -9)$ lies on l_1:

$$7 = 5 - 2\lambda \Rightarrow \lambda = -1, \quad 8 = -2 + 8\lambda \Rightarrow \lambda = \frac{5}{4}, \quad -9 = -1 + 13.5\lambda \Rightarrow \lambda = -\frac{16}{27}.$$

Inconsistent values of λ, so the lines are not coincident. So the lines are parallel.

13)

$$2 + 2\lambda = 4 + 3\mu \Rightarrow 2\lambda - 3\mu = 2 \quad (1), \quad 3 + \lambda = -5 + 2\mu \Rightarrow \lambda - 2\mu = -8 \quad (2),$$

$$-3 + 6\lambda = 3 + 9\mu \Rightarrow 6\lambda - 9\mu = 6 \quad (3).$$

Solve (1) and (2): $\lambda = 28$ and $\mu = 18$. Check in (3): $6(28) - 9(18) = 6$ as required. So the lines intersect. Substitute $\lambda = 28$ into l_1. The lines intersect at $(58, 31, 165)$.

14)

$$4 - 3\lambda = 6 + 8\mu \Rightarrow -3\lambda - 8\mu = 2 \quad (1), \quad -1 + 2\lambda = -2 - \mu \Rightarrow 2\lambda + \mu = -1 \quad (2),$$

$$5 + 10\lambda = 1 + 8\mu \Rightarrow 10\lambda - 8\mu = -4 \quad (3).$$

Solve (1) and (2): $\lambda = -\frac{6}{13}$ and $\mu = -\frac{1}{13}$. Check in (3): $10(-\frac{6}{13}) - 8(-\frac{1}{13}) = -4$ as required. So the lines intersect. Substitute $\lambda = -\frac{6}{13}$ into l_1. The lines intersect at $(\frac{70}{13}, -\frac{25}{13}, \frac{5}{13})$.

15 a)

$$6 = k(-4) \Rightarrow k = -\frac{3}{2}, \quad -9 = k(6) \Rightarrow k = -\frac{3}{2}, \quad 16 = k(-\frac{32}{3}) \Rightarrow k = -\frac{3}{2}.$$

$$\therefore \begin{pmatrix} 6 \\ -9 \\ 16 \end{pmatrix} = -\frac{3}{2} \begin{pmatrix} -4 \\ 6 \\ -\frac{32}{3} \end{pmatrix}.$$

Check if $(-4, 3, -8)$ lies on l_1:

$$-4 = 1 + 6\lambda \Rightarrow \lambda = -\frac{5}{6}, \quad 3 = 9 - 9\lambda \Rightarrow \lambda = \frac{2}{3}, \quad 8 = -2 + 16\lambda \Rightarrow \lambda = \frac{5}{8}.$$

Inconsistent values of λ, so the lines are not coincident. So the lines are parallel.

15 b)

$$3 = k(-4.5) \Rightarrow k = -\frac{2}{3}, \quad -8 = k(11.5) \Rightarrow k = -\frac{16}{23}, \quad 5 = k(-7.5) \Rightarrow k = -\frac{2}{3}.$$

Inconsistent values of k, so the lines are not parallel (or coincident).

$$-5 + 3\lambda = 1 - 4.5\mu \Rightarrow 3\lambda + 4.5\mu = 6 \quad (1),$$

$$-2 - 8\lambda = 1 + 11.5\mu \Rightarrow -8\lambda - 11.5\mu = 3 \quad (2), \quad 7 + 5\lambda = -7 - 7.5\mu \Rightarrow 5\lambda + 7.5\mu = -14 \quad (3).$$

Solve (1) and (2): $\lambda = -55$ and $\mu = 38$. Check in (3): $5(-55) + 7.5(38) = 10 \neq -14$. So the lines do not intersect. l_1 and l_2 are skew lines.

15 c)

$$2 = k(-2) \Rightarrow k = -1, \quad -1 = k(6) \Rightarrow k = -\frac{1}{6}, \quad -8 = k(7) \Rightarrow k = -\frac{8}{7}.$$

Inconsistent values of k, so the lines are not parallel (or coincident).

$$2 + 2\lambda = 4 - 2\mu \Rightarrow 2\lambda + 2\mu = 2 \quad (1), \quad -9 - \lambda = 5 + 6\mu \Rightarrow -\lambda - 6\mu = 14 \quad (2),$$

$$3 - 8\lambda = -8 + 7\mu \Rightarrow -8\lambda - 7\mu = -11 \quad (3).$$

Solve (1) and (2): $\lambda = 4$ and $\mu = -3$. Check in (3): $-8(4) - 7(-3) = -11$ as required. So the lines intersect. Substitute $\lambda = 4$ into l_1. The lines intersect at $(10, -13, -29)$.

16 a)

$$2 + 3\lambda = -3 + \mu \Rightarrow 3\lambda - \mu = -5 \quad (1), \quad -4 + k\lambda = 5 - 6\mu \Rightarrow k\lambda + 6\mu = 9 \quad (2),$$

$$7 + 8\lambda = -9 + 4\mu \Rightarrow 8\lambda - 4\mu = -16 \quad (3).$$

Solve (1) and (3): $\lambda = -1$ and $\mu = 2$. Substitute into (2): $k(-1) + 6(2) = 9 \Rightarrow k = 3$.

16 b)

Substitute $\mu = 2$ into l_2. The lines intersect at $(-1, -7, -1)$, so the position vector is $\begin{pmatrix} -1 \\ -7 \\ -1 \end{pmatrix}$ or $-\mathbf{i} - 7\mathbf{j} - \mathbf{k}$.

17)

$$\begin{pmatrix} 4 \\ -5 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 6 \\ k \\ -1 \end{pmatrix} = 0 \Rightarrow 4 \times 6 + (-5) \times k + 3 \times (-1) = 0$$

$$\Rightarrow 24 - 5k - 3 = 0 \Rightarrow 5k = 21 \Rightarrow k = \frac{21}{5}.$$

18)

$$\begin{pmatrix} 2 \\ -3 \\ 4 \end{pmatrix} \cdot \begin{pmatrix} 24 \\ 4 \\ -9 \end{pmatrix} = 2 \times 24 + (-3) \times 4 + 4 \times (-9) = 48 - 12 - 36 = 0.$$

Hence l_1 and l_2 are perpendicular.

19)

$$\begin{pmatrix} 2 \\ -1 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} = 2 \times 0 + (-1) \times 1 + (-1) \times (-1) = 0 - 1 + 1 = 0.$$

Hence l_1 and l_2 are perpendicular.

20 a)

$$\begin{pmatrix} -2 \\ 3 \\ 5 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -5 \\ -1 \end{pmatrix} = (-2) \times 3 + 3 \times (-5) + 5 \times (-1) = -6 - 15 - 5 = -26 \neq 0.$$

So the vectors are not perpendicular.

20 b)

$$1 - 2\lambda = -13 + 3\mu \Rightarrow 2\lambda + 3\mu = 14 \quad (1), \quad 6 + 3\lambda = 28 - 5\mu \Rightarrow 3\lambda + 5\mu = 22 \quad (2),$$
$$2 + 5\lambda = 24 - \mu \Rightarrow 5\lambda + \mu = 22 \quad (3).$$

Solve (1) and (2): $\lambda = 4$ and $\mu = 2$. Check in (3): $5(4) + 2 = 22$ as required. So the lines intersect. Substitute $\lambda = 4$ into l_1. The lines intersect at $(-7, 18, 22)$.

20 c)

Let $l_3 : \mathbf{r} = \begin{pmatrix} -7 \\ 18 \\ 22 \end{pmatrix} + \gamma \begin{pmatrix} a \\ b \\ c \end{pmatrix}$. l_3 is perpendicular to both l_1 and l_2, so

$$\begin{pmatrix} a \\ b \\ c \end{pmatrix} \cdot \begin{pmatrix} -2 \\ 3 \\ 5 \end{pmatrix} = 0 \Rightarrow -2a + 3b + 5c = 0,$$

$$\begin{pmatrix} a \\ b \\ c \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -5 \\ -1 \end{pmatrix} = 0 \Rightarrow 3a - 5b - c = 0.$$

Let $c = 1$ (for example):

$$-2a + 3b + 5 = 0 \Rightarrow -2a + 3b = -5 \quad (4), \quad 3a - 5b - 1 = 0 \Rightarrow 3a - 5b = 1 \quad (5).$$

Solve (4) and (5): $a = 22$ and $b = 13$.

$$\therefore l_3 : \mathbf{r} = \begin{pmatrix} -7 \\ 18 \\ 22 \end{pmatrix} + \gamma \begin{pmatrix} 22 \\ 13 \\ 1 \end{pmatrix}.$$

21 a)

$$\cos\theta = \frac{\mathbf{d}_1 \cdot \mathbf{d}_2}{|\mathbf{d}_1||\mathbf{d}_2|} = \frac{1 \times 3 + 2 \times (-2) + 1 \times 2}{\sqrt{1^2 + 2^2 + 1^2}\sqrt{3^2 + (-2)^2 + 2^2}} = \frac{1}{\sqrt{6}\sqrt{17}} = \frac{\sqrt{102}}{102}.$$

$$\theta = \arccos(\frac{\sqrt{102}}{102}) = 84.3^\circ \text{ (1 DP)}.$$

21 b)

$$\cos\theta = \frac{\mathbf{d}_1 \cdot \mathbf{d}_2}{|\mathbf{d}_1||\mathbf{d}_2|} = \frac{0\times 4 + 5\times 0 + 2\times 3}{\sqrt{0^2+5^2+2^2}\sqrt{4^2+0^2+3^2}} = \frac{6}{\sqrt{29}\sqrt{25}} = \frac{6\sqrt{29}}{145}.$$

$$\theta = \arccos(\frac{6\sqrt{29}}{145}) = 77.1^\circ \text{ (1 DP)}.$$

21 c)

$$\cos\theta = \frac{\mathbf{d}_1 \cdot \mathbf{d}_2}{|\mathbf{d}_1||\mathbf{d}_2|} = \frac{3\times(-7) + (-2)\times 6 + (-8)\times 6}{\sqrt{3^2+(-2)^2+(-8)^2}\sqrt{(-7)^2+6^2+6^2}} = \frac{-81}{\sqrt{77}\sqrt{121}}.$$

$$\theta = \arccos(-\frac{81\sqrt{77}}{847}) = 147.051\ldots^\circ.$$

Acute angle is $180^\circ - 147.051\ldots^\circ = 32.9^\circ$ (1 DP).

21 d)

$$\cos\theta = \frac{\mathbf{d}_1 \cdot \mathbf{d}_2}{|\mathbf{d}_1||\mathbf{d}_2|} = \frac{4\times(-15) + (-9)\times 3 + 2\times 1}{\sqrt{4^2+(-9)^2+2^2}\sqrt{(-15)^2+3^2+1^2}} = \frac{-85}{\sqrt{101}\sqrt{235}} = -\frac{85}{\sqrt{23735}}.$$

$$\theta = \arccos(-\frac{85}{\sqrt{23735}}) = 123.485\ldots^\circ.$$

Acute angle is $180^\circ - 123.485\ldots^\circ = 56.5^\circ$ (1 DP).

22 a)

Let $\mathbf{d}_1 = \begin{pmatrix} 2 \\ -7 \\ 6 \end{pmatrix}$ and $\mathbf{d}_2 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$.

$$\cos\theta = \frac{\mathbf{d}_1 \cdot \mathbf{d}_2}{|\mathbf{d}_1||\mathbf{d}_2|} = \frac{2\times 1 + (-7)\times 0 + 6\times 0}{\sqrt{2^2+(-7)^2+6^2}\sqrt{1^2+0^2+0^2}} = \frac{2}{\sqrt{89}\sqrt{1}} = \frac{2\sqrt{89}}{89}.$$

$$\theta = \arccos(\frac{2\sqrt{89}}{89}) = 77.8^\circ \text{ (1 DP)}.$$

22 b)

Let $\mathbf{d}_1 = \begin{pmatrix} 2 \\ -7 \\ 6 \end{pmatrix}$ and $\mathbf{d}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$.

$$\cos\theta = \frac{\mathbf{d}_1 \cdot \mathbf{d}_2}{|\mathbf{d}_1||\mathbf{d}_2|} = \frac{2\times 0 + (-7)\times 1 + 6\times 0}{\sqrt{2^2+(-7)^2+6^2}\sqrt{0^2+1^2+0^2}} = -\frac{7}{\sqrt{89}\sqrt{1}} = -\frac{7\sqrt{89}}{89}.$$

$$\theta = \arccos(-\frac{7\sqrt{89}}{89}) = 137.901\ldots^\circ.$$

Acute angle is $180^\circ - 137.901\ldots^\circ = 42.1^\circ$ (1 DP).

22 c)

Let $\mathbf{d}_1 = \begin{pmatrix} 2 \\ -7 \\ 6 \end{pmatrix}$ and $\mathbf{d}_2 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$.

$$\cos\theta = \frac{\mathbf{d}_1 \cdot \mathbf{d}_2}{|\mathbf{d}_1||\mathbf{d}_2|} = \frac{2 \times 0 + (-7) \times 0 + 6 \times 1}{\sqrt{2^2 + (-7)^2 + 6^2}\sqrt{0^2 + 0^2 + 1^2}} = \frac{6}{\sqrt{89}\sqrt{1}} = \frac{6\sqrt{89}}{89}.$$

$$\theta = \arccos(\frac{6\sqrt{89}}{89}) = 50.5^\circ \text{ (1 DP)}.$$

23)

$$\cos\theta = \frac{\mathbf{d}_1 \cdot \mathbf{d}_2}{|\mathbf{d}_1||\mathbf{d}_2|} = \frac{(-2) \times 10 + (-12) \times 1 + 3 \times 5}{\sqrt{(-2)^2 + (-12)^2 + 3^2}\sqrt{10^2 + 1^2 + 5^2}} = \frac{-17}{\sqrt{157}\sqrt{126}} = -\frac{17\sqrt{2198}}{6594}.$$

$$\theta = \arccos(-\frac{17\sqrt{2198}}{6594}) = 96.942\ldots^\circ.$$

Obtuse angle is 96.9° (1 DP).

24)

$$\cos\theta = \frac{\mathbf{d}_1 \cdot \mathbf{d}_2}{|\mathbf{d}_1||\mathbf{d}_2|} = \frac{1 \times 2 + (-1) \times 4 + 1 \times 11}{\sqrt{1^2 + (-1)^2 + 1^2}\sqrt{2^2 + 4^2 + 11^2}} = \frac{9}{\sqrt{3}\sqrt{141}} = \frac{3\sqrt{47}}{47}.$$

$$\theta = \arccos(\frac{3\sqrt{47}}{47}) = 64.049\ldots^\circ.$$

Obtuse angle is $180^\circ - 64.049\ldots^\circ = 116.0^\circ$ (1 DP).

25)

$$\begin{pmatrix} -1 & 2 & 3 \\ 1 & -1 & 2 \\ 0 & 2 & 1 \end{pmatrix} \begin{pmatrix} 3-\lambda \\ -2+6\lambda \\ 5+2\lambda \end{pmatrix} = \begin{pmatrix} -(3-\lambda) + 2(-2+6\lambda) + 3(5+2\lambda) \\ (3-\lambda) - (-2+6\lambda) + 2(5+2\lambda) \\ 0(3-\lambda) + 2(-2+6\lambda) + (5+2\lambda) \end{pmatrix}$$

$$= \begin{pmatrix} -3+\lambda-4+12\lambda+15+6\lambda \\ 3-\lambda+2-12\lambda+10+4\lambda \\ -4+12\lambda+5+2\lambda \end{pmatrix} = \begin{pmatrix} 8+19\lambda \\ 15-9\lambda \\ 1+14\lambda \end{pmatrix} = \begin{pmatrix} 8 \\ 15 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 19 \\ -9 \\ 14 \end{pmatrix}.$$

Therefore $l_2 : \mathbf{r} = \begin{pmatrix} 8 \\ 15 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 19 \\ -9 \\ 14 \end{pmatrix}$.

26 a)

$$\begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 7+2\lambda \\ -1+5\lambda \\ 3-6\lambda \end{pmatrix} = \begin{pmatrix} 2(7+2\lambda) + 0(-1+5\lambda) + 0(3-6\lambda) \\ 0(7+2\lambda) + 1(-1+5\lambda) + 0(3-6\lambda) \\ 0(7+2\lambda) + 0(-1+5\lambda) + 1(3-6\lambda) \end{pmatrix}$$

$$= \begin{pmatrix} 14+4\lambda \\ -1+5\lambda \\ 3-6\lambda \end{pmatrix} = \begin{pmatrix} 14 \\ -1 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ 5 \\ -6 \end{pmatrix}.$$

Therefore $l_2 : \mathbf{r} = \begin{pmatrix} 14 \\ -1 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ 5 \\ -6 \end{pmatrix}$.

26 b)

$$\begin{pmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{pmatrix} \begin{pmatrix} 7+2\lambda \\ -1+5\lambda \\ 3-6\lambda \end{pmatrix} = \begin{pmatrix} 3(7+2\lambda)+0(-1+5\lambda)+0(3-6\lambda) \\ 0(7+2\lambda)+3(-1+5\lambda)+0(3-6\lambda) \\ 0(7+2\lambda)+0(-1+5\lambda)+3(3-6\lambda) \end{pmatrix}$$

$$= \begin{pmatrix} 21+6\lambda \\ -3+15\lambda \\ 9-18\lambda \end{pmatrix} = \begin{pmatrix} 21 \\ -3 \\ 9 \end{pmatrix} + \lambda \begin{pmatrix} 6 \\ 15 \\ -18 \end{pmatrix}.$$

Therefore $l_2 : \mathbf{r} = \begin{pmatrix} 21 \\ -3 \\ 9 \end{pmatrix} + \lambda \begin{pmatrix} 6 \\ 15 \\ -18 \end{pmatrix}$ or $l_2 : \mathbf{r} = \begin{pmatrix} 21 \\ -3 \\ 9 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 5 \\ -6 \end{pmatrix}$.

26 c)

$$\begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 7+2\lambda \\ -1+5\lambda \\ 3-6\lambda \end{pmatrix} = \begin{pmatrix} -1(7+2\lambda)+0(-1+5\lambda)+0(3-6\lambda) \\ 0(7+2\lambda)+1(-1+5\lambda)+0(3-6\lambda) \\ 0(7+2\lambda)+0(-1+5\lambda)+1(3-6\lambda) \end{pmatrix}$$

$$= \begin{pmatrix} -7-2\lambda \\ -1+5\lambda \\ 3-6\lambda \end{pmatrix} = \begin{pmatrix} -7 \\ -1 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 5 \\ -6 \end{pmatrix}.$$

Therefore $l_2 : \mathbf{r} = \begin{pmatrix} -7 \\ -1 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 5 \\ -6 \end{pmatrix}$.

26 d)

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 7+2\lambda \\ -1+5\lambda \\ 3-6\lambda \end{pmatrix} = \begin{pmatrix} 1(7+2\lambda)+0(-1+5\lambda)+0(3-6\lambda) \\ 0(7+2\lambda)-1(-1+5\lambda)+0(3-6\lambda) \\ 0(7+2\lambda)+0(-1+5\lambda)+1(3-6\lambda) \end{pmatrix}$$

$$= \begin{pmatrix} 7+2\lambda \\ 1-5\lambda \\ 3-6\lambda \end{pmatrix} = \begin{pmatrix} 7 \\ 1 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -5 \\ -6 \end{pmatrix}.$$

Therefore $l_2 : \mathbf{r} = \begin{pmatrix} 7 \\ 1 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -5 \\ -6 \end{pmatrix}$.

26 e)

$$\begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 7+2\lambda \\ -1+5\lambda \\ 3-6\lambda \end{pmatrix} = \begin{pmatrix} 0(7+2\lambda)+0(-1+5\lambda)+1(3-6\lambda) \\ 0(7+2\lambda)+1(-1+5\lambda)+0(3-6\lambda) \\ -(7+2\lambda)+0(-1+5\lambda)+0(3-6\lambda) \end{pmatrix}$$

$$= \begin{pmatrix} 3-6\lambda \\ -1+5\lambda \\ -7-2\lambda \end{pmatrix} = \begin{pmatrix} 3 \\ -1 \\ -7 \end{pmatrix} + \lambda \begin{pmatrix} -6 \\ 5 \\ -2 \end{pmatrix}.$$

Therefore $l_2 : \mathbf{r} = \begin{pmatrix} 3 \\ -1 \\ -7 \end{pmatrix} + \lambda \begin{pmatrix} -6 \\ 5 \\ -2 \end{pmatrix}$.

26 f)

$$\begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 7+2\lambda \\ -1+5\lambda \\ 3-6\lambda \end{pmatrix} = \begin{pmatrix} 0(7+2\lambda) - (-1+5\lambda) + 0(3-6\lambda) \\ 1(7+2\lambda) + 0(-1+5\lambda) + 0(3-6\lambda) \\ 0(7+2\lambda) + 0(-1+5\lambda) + 1(3-6\lambda) \end{pmatrix}$$

$$= \begin{pmatrix} 1-5\lambda \\ 7+2\lambda \\ 3-6\lambda \end{pmatrix} = \begin{pmatrix} 1 \\ 7 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} -5 \\ 2 \\ -6 \end{pmatrix}.$$

Therefore $l_2 : \mathbf{r} = \begin{pmatrix} 1 \\ 7 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} -5 \\ 2 \\ -6 \end{pmatrix}$.

27 a)

$$\begin{pmatrix} 1 & 2 & 0 \\ 0 & 3 & 1 \\ 2 & 1 & 1 \end{pmatrix} \begin{pmatrix} 5 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 9 \\ 7 \\ 13 \end{pmatrix}.$$

Let $\mathbf{d}_1 = \begin{pmatrix} 5 \\ 2 \\ 1 \end{pmatrix}$ and $\mathbf{d}_2 = \begin{pmatrix} 9 \\ 7 \\ 13 \end{pmatrix}$.

$$\cos\theta = \frac{\mathbf{d}_1 \cdot \mathbf{d}_2}{|\mathbf{d}_1||\mathbf{d}_2|} = \frac{5 \times 9 + 2 \times 7 + 1 \times 13}{\sqrt{5^2+2^2+1^2}\sqrt{9^2+7^2+13^2}} = \frac{72}{\sqrt{30}\sqrt{299}} = \frac{12\sqrt{8970}}{1495}.$$

$$\theta = \arccos(\frac{12\sqrt{8970}}{1495}) = 40.5^\circ \text{ (1 DP)}.$$

27 b)

$$\begin{pmatrix} -2 & 3 & 1 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{pmatrix} \begin{pmatrix} 5 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} -3 \\ 12 \\ 20 \end{pmatrix}.$$

Let $\mathbf{d}_1 = \begin{pmatrix} 5 \\ 2 \\ 1 \end{pmatrix}$ and $\mathbf{d}_2 = \begin{pmatrix} -3 \\ 12 \\ 20 \end{pmatrix}$.

$$\cos\theta = \frac{\mathbf{d}_1 \cdot \mathbf{d}_2}{|\mathbf{d}_1||\mathbf{d}_2|} = \frac{5 \times (-3) + 2 \times 12 + 1 \times 20}{\sqrt{5^2+2^2+1^2}\sqrt{(-3)^2+12^2+20^2}} = \frac{29}{\sqrt{30}\sqrt{553}} = \frac{29\sqrt{16590}}{16590}.$$

$$\theta = \arccos(\frac{29\sqrt{16590}}{16590}) = 77.0^\circ \text{ (1 DP)}.$$

27 c)

$$\begin{pmatrix} 5 & -3 & -4 \\ 2 & 8 & -3 \\ -5 & -4 & -1 \end{pmatrix} \begin{pmatrix} 5 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 15 \\ 23 \\ -34 \end{pmatrix}.$$

Let $\mathbf{d}_1 = \begin{pmatrix} 5 \\ 2 \\ 1 \end{pmatrix}$ and $\mathbf{d}_2 = \begin{pmatrix} 15 \\ 23 \\ -34 \end{pmatrix}$.

$$\cos\theta = \frac{\mathbf{d}_1 \cdot \mathbf{d}_2}{|\mathbf{d}_1||\mathbf{d}_2|} = \frac{5 \times 15 + 2 \times 23 + 1 \times (-34)}{\sqrt{5^2 + 2^2 + 1^2}\sqrt{15^2 + 23^2 + (-34)^2}} = \frac{87}{\sqrt{30}\sqrt{1910}} = \frac{29\sqrt{573}}{1910}.$$

$$\theta = \arccos(\frac{29\sqrt{573}}{1910}) = 68.7^\circ \text{ (1 DP)}.$$

28 a)

$(2 + 4\lambda, -3 + 3\lambda, 1 + \lambda)$.

28 b)

$(5, -9 - 3\mu, 16\mu)$.

29 a)

$$(-4 + (1) \times 3, 2 + (1) \times 8, 5 + (1) \times (-4)) = (-1, 10, 1).$$

29 b)

$$(-4 + (-1) \times 3, 2 + (-1) \times 8, 5 + (-1) \times (-4)) = (-7, -6, 9).$$

29 c)

$$(-4 + (-4) \times 3, 2 + (-4) \times 8, 5 + (-4) \times (-4)) = (-16, -30, 21).$$

30)

$$\overrightarrow{OA} = \begin{pmatrix} 2 \\ 1 \\ -5 \end{pmatrix} \quad \text{and} \quad \overrightarrow{OB} = \begin{pmatrix} 2 + 4\lambda \\ 2 - \lambda \\ -7 + \lambda \end{pmatrix}.$$

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} 2 + 4\lambda \\ 2 - \lambda \\ -7 + \lambda \end{pmatrix} - \begin{pmatrix} 2 \\ 1 \\ -5 \end{pmatrix} = \begin{pmatrix} 4\lambda \\ 1 - \lambda \\ -2 + \lambda \end{pmatrix}.$$

31)

$$\overrightarrow{OP} = \begin{pmatrix} 3 \\ -7 \\ 2 \end{pmatrix} \quad \text{and} \quad \overrightarrow{OQ} = \begin{pmatrix} 4-3\lambda \\ -1+2\lambda \\ 2+5\lambda \end{pmatrix}.$$

$$\overrightarrow{QP} = \overrightarrow{OP} - \overrightarrow{OQ} = \begin{pmatrix} 3 \\ -7 \\ 2 \end{pmatrix} - \begin{pmatrix} 4-3\lambda \\ -1+2\lambda \\ 2+5\lambda \end{pmatrix} = \begin{pmatrix} 3\lambda - 1 \\ -6-2\lambda \\ -5\lambda \end{pmatrix}.$$

32 a)

$$\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 2-\lambda \\ 1+5\lambda \\ 2\lambda - 1 \end{pmatrix} = 0$$

$$\Rightarrow 1\times(2-\lambda)+1\times(1+5\lambda)+1\times(2\lambda-1)=0 \Rightarrow 2-\lambda+1+5\lambda+2\lambda-1=0 \Rightarrow 6\lambda+2=0 \Rightarrow \lambda=-\frac{1}{3}.$$

32 b)

$$\begin{pmatrix} 2 \\ 0 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 2-\lambda \\ 1+5\lambda \\ 2\lambda - 1 \end{pmatrix} = 0$$

$$\Rightarrow 2\times(2-\lambda)+0\times(1+5\lambda)+3\times(2\lambda-1)=0 \Rightarrow 4-2\lambda+6\lambda-3=0 \Rightarrow 4\lambda+1=0 \Rightarrow \lambda=-\frac{1}{4}.$$

32 c)

$$\begin{pmatrix} -1 \\ 3 \\ -5 \end{pmatrix} \cdot \begin{pmatrix} 2-\lambda \\ 1+5\lambda \\ 2\lambda - 1 \end{pmatrix} = 0$$

$$\begin{aligned} \Rightarrow (-1)\times(2-\lambda)+3\times(1+5\lambda)+(-5)\times(2\lambda-1)&=0 \\ \Rightarrow -2+\lambda+3+15\lambda-10\lambda+5&=0 \\ \Rightarrow 6\lambda+6&=0 \\ \Rightarrow \lambda&=-1. \end{aligned}$$

33 a)

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} 2 \\ 5 \\ -3 \end{pmatrix} - \begin{pmatrix} 3 \\ 7 \\ -1 \end{pmatrix} = \begin{pmatrix} -1 \\ -2 \\ -2 \end{pmatrix}.$$

$$\therefore l_1 : \mathbf{r} = \begin{pmatrix} 3 \\ 7 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ -2 \\ -2 \end{pmatrix}.$$

33 b)

$$38 = 2 + 6\mu \Rightarrow \mu = 6, \quad -17 = -5 - 2\mu \Rightarrow \mu = 6, \quad -15 = 3 - 3\mu \Rightarrow \mu = 6.$$

Consistent values of μ, therefore $P(38, -17, -15)$ lies on l_2.

33 c)

As C lies on l_1, let $\overrightarrow{OC} = \begin{pmatrix} 3-\lambda \\ 7-2\lambda \\ -1-2\lambda \end{pmatrix}$.

$$\overrightarrow{CP} = \overrightarrow{OP} - \overrightarrow{OC} = \begin{pmatrix} 38 \\ -17 \\ -15 \end{pmatrix} - \begin{pmatrix} 3-\lambda \\ 7-2\lambda \\ -1-2\lambda \end{pmatrix} = \begin{pmatrix} \lambda+35 \\ 2\lambda-24 \\ 2\lambda-14 \end{pmatrix}.$$

If $\overrightarrow{CP}$ is perpendicular to l_2 then

$$\begin{pmatrix} \lambda+35 \\ 2\lambda-24 \\ 2\lambda-14 \end{pmatrix} \cdot \begin{pmatrix} 6 \\ -2 \\ -3 \end{pmatrix} = 0$$

$\Rightarrow 6(\lambda+35)-2(2\lambda-24)-3(2\lambda-14) = 0 \Rightarrow 6\lambda+210-4\lambda+48-6\lambda+42 = 0 \Rightarrow -4\lambda+300 = 0 \Rightarrow \lambda = 75.$

$$\therefore \overrightarrow{OC} = \begin{pmatrix} 3-(75) \\ 7-2(75) \\ -1-2(75) \end{pmatrix} = \begin{pmatrix} -72 \\ -143 \\ -151 \end{pmatrix}.$$

So $C(-72, -143, -151)$.

34 a)

$$\overrightarrow{OA} = \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}.$$

$$\overrightarrow{OC} = 99\overrightarrow{OA} = 99 \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix} = \begin{pmatrix} 198 \\ -99 \\ 198 \end{pmatrix}.$$

Therefore $C(198, -99, 198)$.

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} 3 \\ 8 \\ -2 \end{pmatrix} - \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ 9 \\ -4 \end{pmatrix}.$$

$$\therefore l : \mathbf{r} = \begin{pmatrix} 198 \\ -99 \\ 198 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 9 \\ -4 \end{pmatrix}.$$

34 b)

$$\overrightarrow{OD} = \begin{pmatrix} 198+\lambda \\ -99+9\lambda \\ 198-4\lambda \end{pmatrix}.$$

$$\overrightarrow{BD} = \overrightarrow{OD} - \overrightarrow{OB} = \begin{pmatrix} 198+\lambda \\ -99+9\lambda \\ 198-4\lambda \end{pmatrix} - \begin{pmatrix} 3 \\ 8 \\ -2 \end{pmatrix} = \begin{pmatrix} 195+\lambda \\ -107+9\lambda \\ 200-4\lambda \end{pmatrix}.$$

If angle $ABD = 90°$, then $\overrightarrow{AB} \cdot \overrightarrow{BD} = 0$:

$$\begin{pmatrix} 1 \\ 9 \\ -4 \end{pmatrix} \cdot \begin{pmatrix} 195+\lambda \\ -107+9\lambda \\ 200-4\lambda \end{pmatrix} = 0$$

$$\begin{aligned} \Rightarrow 1(195+\lambda) + 9(-107+9\lambda) - 4(200-4\lambda) &= 0 \\ \Rightarrow 195+\lambda-963+81\lambda-800+16\lambda &= 0 \\ \Rightarrow 98\lambda - 1568 &= 0 \\ \Rightarrow \lambda &= 16. \end{aligned}$$

$$\therefore \overrightarrow{OD} = \begin{pmatrix} 198+(16) \\ -99+9(16) \\ 198-4(16) \end{pmatrix} = \begin{pmatrix} 214 \\ 45 \\ 134 \end{pmatrix}.$$

So $D(214, 45, 134)$.

35)

l_1 and l_2 intersect:

$$10.25 - 3\lambda = a + 2\mu \quad (1), \qquad -6 + 4\lambda = 1 - \mu \quad (2), \qquad 5 = 7 + b\mu \quad (3).$$

l_1 and l_2 form an angle of $120°$:

$$\begin{aligned} \cos 120° &= \frac{(-3)\times 2 + 4\times(-1) + 0\times b}{\sqrt{(-3)^2+4^2+0^2}\sqrt{2^2+(-1)^2+b^2}} \\ \Rightarrow -\frac{1}{2} &= \frac{-10}{\sqrt{25}\sqrt{5+b^2}} \\ \Rightarrow \sqrt{5+b^2} &= 4 \\ \Rightarrow 5+b^2 &= 16 \\ \Rightarrow b^2 &= 11 \\ \Rightarrow b &= \sqrt{11} \text{ as } b \text{ is a positive constant.} \end{aligned}$$

Solve (3): $5 = 7 + \sqrt{11}\mu \Rightarrow \mu = -\frac{2}{\sqrt{11}} = -\frac{2\sqrt{11}}{11}$. Substitute into (2):

$$-6 + 4\lambda = 1 - (-\frac{2\sqrt{11}}{11}) \Rightarrow \lambda = \frac{77+2\sqrt{11}}{44}.$$

Substitute into (1):

$$10.25 - 3(\frac{77+2\sqrt{11}}{44}) = a + 2(-\frac{2\sqrt{11}}{11}) \Rightarrow a = \frac{110+5\sqrt{11}}{22} = 5 + \frac{5\sqrt{11}}{22}.$$

36 a)

$$\overrightarrow{OA} = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} \quad \text{and} \quad \overrightarrow{OB} = \begin{pmatrix} 3 \\ -1 \\ -4 \end{pmatrix}.$$

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} 3 \\ -1 \\ -4 \end{pmatrix} - \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \\ -5 \end{pmatrix}.$$

$$l : \mathbf{r} = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -2 \\ -5 \end{pmatrix}.$$

36 b)

$$\overrightarrow{OC} = \begin{pmatrix} a \\ 1 \\ -1 \end{pmatrix}, \quad \overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA} = \begin{pmatrix} a-2 \\ 0 \\ -2 \end{pmatrix}.$$

AC is perpendicular to l, so

$$\begin{pmatrix} a-2 \\ 0 \\ -2 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -2 \\ -5 \end{pmatrix} = 0$$

$$\Rightarrow (a-2) \times 1 + 0 \times (-2) + (-2) \times (-5) = 0$$

$$\Rightarrow a - 2 + 0 + 10 = 0$$

$$\Rightarrow a = -8.$$

36 c)

$$\overrightarrow{AC} = \begin{pmatrix} -8-2 \\ 0 \\ -2 \end{pmatrix} = \begin{pmatrix} -10 \\ 0 \\ -2 \end{pmatrix}, \qquad \overrightarrow{AB} = \begin{pmatrix} 1 \\ -2 \\ -5 \end{pmatrix}.$$

$$|\overrightarrow{AC}| = \sqrt{(-10)^2 + 0^2 + (-2)^2} = 2\sqrt{26},$$

$$|\overrightarrow{AB}| = \sqrt{1^2 + (-2)^2 + (-5)^2} = \sqrt{30}.$$

$$\text{Area of triangle} = \frac{1}{2}|\overrightarrow{AC}||\overrightarrow{AB}|$$

$$= \frac{1}{2} \times 2\sqrt{26} \times \sqrt{30}$$

$$= 2\sqrt{195} \text{ units}^2.$$

37 a)

$$\overrightarrow{OA} = \begin{pmatrix} 3 \\ -5 \\ -6 \end{pmatrix} - 2 \begin{pmatrix} 2 \\ 8 \\ -9 \end{pmatrix} = \begin{pmatrix} -1 \\ -21 \\ 12 \end{pmatrix}.$$

So $A(-1, -21, 12)$.

37 b)

Let $\begin{pmatrix} a \\ b \\ c \end{pmatrix}$ be a direction vector that is perpendicular to $\begin{pmatrix} 2 \\ 8 \\ -9 \end{pmatrix}$.

$$\begin{pmatrix} a \\ b \\ c \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 8 \\ -9 \end{pmatrix} = 0 \Rightarrow 2a + 8b - 9c = 0.$$

Let $a = 0, b = 9, c = 8$. Then $2 \times 0 + 8 \times 9 - 9 \times 8 = 0$ as required. So the vector equation of a line that is perpendicular to l and passes through A could be $\mathbf{r} = \begin{pmatrix} -1 \\ -21 \\ 12 \end{pmatrix} + \mu \begin{pmatrix} 0 \\ 9 \\ 8 \end{pmatrix}$.

38 a)

Note that $\begin{pmatrix} -4 \\ 1 \\ 5 \end{pmatrix} = -\begin{pmatrix} 4 \\ -1 \\ -5 \end{pmatrix}$ so l_1 and l_3 are parallel, and that $\begin{pmatrix} -2 \\ 0 \\ 1 \end{pmatrix} = \frac{1}{3}\begin{pmatrix} -6 \\ 0 \\ 3 \end{pmatrix}$ so l_2 and l_4 are parallel. So the quadrilateral has two pairs of parallel sides, and so is either a parallelogram or a rhombus. Find the intersection of l_1 and l_2:

$$-7 - 4\alpha = 3 - 2\beta \Rightarrow -4\alpha + 2\beta = 10 \quad (1), \quad 3 + \alpha = 2 \Rightarrow \alpha = -1 \quad (2),$$

$$10 + 5\alpha = 2 + \beta \Rightarrow 5\alpha - \beta = -8 \quad (3).$$

Solve (1) and (2): $\alpha = -1$ and $\beta = 3$. Check in (3): $5(-1) - (3) = -8$ as required. Substitute $\alpha = -1$ into l_1:

$$\mathbf{r} = \begin{pmatrix} -7 \\ 3 \\ 10 \end{pmatrix} + (-1)\begin{pmatrix} -4 \\ 1 \\ 5 \end{pmatrix} = \begin{pmatrix} -3 \\ 2 \\ 5 \end{pmatrix}.$$

So the first vertex has coordinates $(-3, 2, 5)$. Find the intersection of l_1 and l_4:

$$-7 - 4\alpha = 19 - 6\delta \Rightarrow -4\alpha + 6\delta = 26 \quad (1), \quad 3 + \alpha = 1 \Rightarrow \alpha = -2 \quad (2),$$

$$10 + 5\alpha = -9 + 3\delta \Rightarrow 5\alpha - 3\delta = -19 \quad (3).$$

Solve (1) and (2): $\alpha = -2$ and $\delta = 3$. Check in (3): $5(-2) - 3(3) = -19$ as required. Substitute $\alpha = -2$ into l_1:

$$\mathbf{r} = \begin{pmatrix} -7 \\ 3 \\ 10 \end{pmatrix} + (-2)\begin{pmatrix} -4 \\ 1 \\ 5 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}.$$

So the second vertex has coordinates $(1, 1, 0)$. Find the intersection of l_2 and l_3:

$$3 - 2\beta = -21 + 4\gamma \Rightarrow -2\beta - 4\gamma = -24 \quad (1), \quad 2 = 5 - \gamma \Rightarrow \gamma = 3 \quad (2),$$

$$2 + \beta = 23 - 5\gamma \Rightarrow \beta + 5\gamma = 21 \quad (3).$$

Solve (1) and (2): $\beta = 6$ and $\gamma = 3$. Check in (3): $(6) + 5(3) = 21$ as required. Substitute $\beta = 6$ into l_2:

$$\mathbf{r} = \begin{pmatrix} 3 \\ 2 \\ 2 \end{pmatrix} + (6)\begin{pmatrix} -2 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -9 \\ 2 \\ 8 \end{pmatrix}.$$

So the third vertex has coordinates $(-9, 2, 8)$. Now let $A(-3, 2, 5)$, $B(1, 1, 0)$ and $C(-9, 2, 8)$ be three of the four vertices of the parallelogram.

$$AB = \sqrt{(1+3)^2 + (1-2)^2 + (0-5)^2} = \sqrt{42}, \quad AC = \sqrt{(-9+3)^2 + (2-2)^2 + (8-5)^2} = \sqrt{45}.$$

$AB \neq AC$, so it is not a rhombus.

38 b)

$$\cos\theta = \frac{\begin{pmatrix}-4\\1\\5\end{pmatrix}\cdot\begin{pmatrix}-2\\0\\1\end{pmatrix}}{|\begin{pmatrix}-4\\1\\5\end{pmatrix}||\begin{pmatrix}-2\\0\\1\end{pmatrix}|} = \frac{(-4)\times(-2)+1\times 0+5\times 1}{\sqrt{(-4)^2+1^2+5^2}\sqrt{(-2)^2+0^2+1^2}} = \frac{13}{\sqrt{42}\sqrt{5}} = \frac{13}{\sqrt{210}}.$$

$$\sin\theta = \frac{\sqrt{(\sqrt{210})^2-13^2}}{\sqrt{210}} = \frac{\sqrt{41}}{\sqrt{210}}.$$

$$\text{Area of triangle} = \frac{1}{2}\times\sqrt{42}\times\sqrt{45}\times\frac{\sqrt{41}}{\sqrt{210}} = \frac{3\sqrt{41}}{2}\ \text{units}^2.$$

So area of parallelogram is $3\sqrt{41}$ units^2.

39 a)

$$\overrightarrow{OA} = \begin{pmatrix}6\\-6\\1\end{pmatrix},\quad \overrightarrow{OB} = \begin{pmatrix}14\\-12\\5\end{pmatrix},\quad \overrightarrow{AB} = \overrightarrow{OB}-\overrightarrow{OA} = \begin{pmatrix}8\\-6\\4\end{pmatrix}.$$

$$l_1 : \mathbf{r} = \begin{pmatrix}6\\-6\\1\end{pmatrix} + \lambda\begin{pmatrix}8\\-6\\4\end{pmatrix}.$$

39 b)

$$6+8\lambda = 0+3\mu \Rightarrow 8\lambda-3\mu = -6 \quad (1),\quad -6-6\lambda = -2-2\mu \Rightarrow -6\lambda+2\mu = 4 \quad (2),$$
$$1+4\lambda = 3-\mu \Rightarrow 4\lambda+\mu = 2 \quad (3).$$

Solve (1) and (2): $\lambda = 0$ and $\mu = 2$. Check in (3): $4(0)+(2) = 2$ as required. So the lines intersect.

39 c)

$$D(0+(4)\times 3, -2+(4)\times(-2), 3+(4)\times(-1)) = D(12,-10,-1).$$

39 d)

The line through D and C is parallel to l_1 and so has the equation

$$\mathbf{r} = \begin{pmatrix}12\\-10\\-1\end{pmatrix} + \alpha\begin{pmatrix}8\\-6\\4\end{pmatrix}.$$

So C has coordinates $(12+8\alpha, -10-6\alpha, -1+4\alpha)$ for some α.

$$\overrightarrow{AD} = \overrightarrow{OD}-\overrightarrow{OA} = \begin{pmatrix}12\\-10\\-1\end{pmatrix} - \begin{pmatrix}6\\-6\\1\end{pmatrix} = \begin{pmatrix}6\\-4\\-2\end{pmatrix}.$$

$$|\overrightarrow{AD}| = \sqrt{6^2+(-4)^2+(-2)^2} = \sqrt{56} = 2\sqrt{14}.$$

$$\overrightarrow{BC} = \overrightarrow{OC} - \overrightarrow{OB} = \begin{pmatrix} 12+8\alpha \\ -10-6\alpha \\ -1+4\alpha \end{pmatrix} - \begin{pmatrix} 14 \\ -12 \\ 5 \end{pmatrix} = \begin{pmatrix} 8\alpha-2 \\ 2-6\alpha \\ 4\alpha-6 \end{pmatrix}.$$

$$|\overrightarrow{BC}| = |\overrightarrow{AD}| = \sqrt{56} \Rightarrow \sqrt{(8\alpha-2)^2+(2-6\alpha)^2+(4\alpha-6)^2} = \sqrt{56}.$$

$$\Rightarrow 64\alpha^2 - 32\alpha + 4 + 4 - 24\alpha + 36\alpha^2 + 16\alpha^2 - 48\alpha + 36 = 56$$

$$\Rightarrow 64\alpha^2 - 32\alpha + 4 + 4 - 24\alpha + 36\alpha^2 + 16\alpha^2 - 48\alpha + 36 = 56 \Rightarrow 116\alpha^2 - 104\alpha - 12 = 0$$

$$\Rightarrow 29\alpha^2 - 26\alpha - 3 = 0 \Rightarrow (29\alpha+3)(\alpha-1) = 0 \Rightarrow \alpha = -\frac{3}{29} \text{ or } \alpha = 1.$$

So either

$$C(12+8(-\frac{3}{29}), -10-6(-\frac{3}{29}), -1+4(-\frac{3}{29})) = C(\frac{324}{29}, -\frac{272}{29}, -\frac{41}{29}),$$

or $C(12+8(1), -10-6(1), -1+4(1)) = C(20, -16, 3)$. If $\alpha = -\frac{3}{29}$ then

$$\overrightarrow{BC} = \begin{pmatrix} 8(-\frac{3}{29})-2 \\ 2-6(-\frac{3}{29}) \\ 4(-\frac{3}{29})-6 \end{pmatrix} = -\frac{2}{29}\begin{pmatrix} 41 \\ -38 \\ 93 \end{pmatrix}.$$

If $\alpha = 1$ then $\overrightarrow{BC} = \begin{pmatrix} 8(1)-2 \\ 2-6(1) \\ 4(1)-6 \end{pmatrix} = \begin{pmatrix} 6 \\ -4 \\ -2 \end{pmatrix} = \overrightarrow{AD}$. So if $\alpha = 1$, then $ABCD$ is a parallelogram and not a trapezium. So $\alpha = -\frac{3}{29}$ and hence $C(\frac{324}{29}, -\frac{272}{29}, -\frac{41}{29})$.

40 a)

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} 1 \\ 4 \\ -6 \end{pmatrix} - \begin{pmatrix} 4 \\ 7 \\ -6 \end{pmatrix} = \begin{pmatrix} -3 \\ -3 \\ 0 \end{pmatrix}.$$

$$\therefore l_1 : \mathbf{r} = \begin{pmatrix} 4 \\ 7 \\ -6 \end{pmatrix} + \lambda \begin{pmatrix} -3 \\ -3 \\ 0 \end{pmatrix}.$$

$$\overrightarrow{CD} = \overrightarrow{OD} - \overrightarrow{OC} = \begin{pmatrix} 3 \\ 3 \\ -9 \end{pmatrix} - \begin{pmatrix} 3 \\ 5 \\ -7 \end{pmatrix} = \begin{pmatrix} 0 \\ -2 \\ -2 \end{pmatrix}.$$

$$\therefore l_2 : \mathbf{r} = \begin{pmatrix} 3 \\ 5 \\ -7 \end{pmatrix} + \mu \begin{pmatrix} 0 \\ -2 \\ -2 \end{pmatrix}.$$

$$4 - 3\lambda = 3 \quad (1), \quad 7 - 3\lambda = 5 - 2\mu \Rightarrow -3\lambda + 2\mu = -2 \quad (2), \quad -6 = -7 - 2\mu \quad (3).$$

Solve (1) and (3): $\lambda = \frac{1}{3}$ and $\mu = -\frac{1}{2}$. Check in (2): $-3(\frac{1}{3}) + 2(-\frac{1}{2}) = -2$ as required. So the lines intersect. Substitute $\lambda = \frac{1}{3}$ into l_1. So P has coordinates $(3, 6, -6)$.

40 b)

$$\overrightarrow{BP} = \overrightarrow{OP} - \overrightarrow{OB} = \begin{pmatrix} 3 \\ 6 \\ -6 \end{pmatrix} - \begin{pmatrix} 1 \\ 4 \\ -6 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ 0 \end{pmatrix}.$$

$$|\overrightarrow{BP}| = \sqrt{2^2+2^2+0^2} = \sqrt{8} = 2\sqrt{2}.$$

E lies on l_2 so it has coordinates $(3, 5-2\mu, -7-2\mu)$ for some value μ.

$$\overrightarrow{PE} = \overrightarrow{OE} - \overrightarrow{OP} = \begin{pmatrix} 3 \\ 5-2\mu \\ -7-2\mu \end{pmatrix} - \begin{pmatrix} 3 \\ 6 \\ -6 \end{pmatrix} = \begin{pmatrix} 0 \\ -1-2\mu \\ -1-2\mu \end{pmatrix}.$$

$$|\overrightarrow{PE}| = \sqrt{8} \Rightarrow \sqrt{0^2 + (-1-2\mu)^2 + (-1-2\mu)^2} = \sqrt{8}$$

$$\Rightarrow 0 + 1 + 4\mu + 4\mu^2 + 1 + 4\mu + 4\mu^2 = 8 \Rightarrow 8\mu^2 + 8\mu - 6 = 0 \Rightarrow 4\mu^2 + 4\mu - 3 = 0$$

$$\Rightarrow (2\mu + 3)(2\mu - 1) = 0 \Rightarrow \mu = -\frac{3}{2} \text{ or } \mu = \frac{1}{2}.$$

If $E = (3, 5-2(-\frac{3}{2}), -7-2(-\frac{3}{2})) = (3, 8, -4)$. Or $E = (3, 5-2(\frac{1}{2}), -7-2(\frac{1}{2})) = (3, 4, -8)$.

41)

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} 4+\sqrt{6} \\ -5-3\sqrt{6} \\ -2-\sqrt{6} \end{pmatrix} - \begin{pmatrix} 4 \\ -5 \\ 4-\sqrt{6} \end{pmatrix} = \begin{pmatrix} \sqrt{6} \\ -3\sqrt{6} \\ -6 \end{pmatrix}.$$

$$\therefore l_1 : \mathbf{r} = \begin{pmatrix} 4 \\ -5 \\ 4-\sqrt{6} \end{pmatrix} + \lambda \begin{pmatrix} \sqrt{6} \\ -3\sqrt{6} \\ -6 \end{pmatrix}.$$

$$\overrightarrow{CD} = \overrightarrow{OD} - \overrightarrow{OC} = \begin{pmatrix} \sqrt{6}-9 \\ 1-6\sqrt{6} \\ 5-2\sqrt{6} \end{pmatrix} - \begin{pmatrix} 3+\sqrt{6} \\ 1 \\ 5 \end{pmatrix} = \begin{pmatrix} -12 \\ -6\sqrt{6} \\ -2\sqrt{6} \end{pmatrix}.$$

$$\therefore l_2 : \mathbf{r} = \begin{pmatrix} 3+\sqrt{6} \\ 1 \\ 5 \end{pmatrix} + \mu \begin{pmatrix} -12 \\ -6\sqrt{6} \\ -2\sqrt{6} \end{pmatrix}.$$

$$4 + \sqrt{6}\lambda = 3 + \sqrt{6} - 12\mu \Rightarrow \sqrt{6}\lambda + 12\mu = -1 + \sqrt{6} \quad (1),$$

$$-5 - 3\sqrt{6}\lambda = 1 - 6\sqrt{6}\mu \Rightarrow -3\sqrt{6}\lambda + 6\sqrt{6}\mu = 6 \quad (2),$$

$$4 - \sqrt{6} - 6\lambda = 5 - 2\sqrt{6}\mu \Rightarrow -6\lambda + 2\sqrt{6}\mu = 1 + \sqrt{6} \quad (3).$$

Solve (1) and (2): $\lambda = -\frac{\sqrt{6}}{6}$ and $\mu = \frac{\sqrt{6}}{12}$. Check in (3): $-6(-\frac{\sqrt{6}}{6}) + 2\sqrt{6}(\frac{\sqrt{6}}{12}) = 1 + \sqrt{6}$ as required. So the lines intersect. Substitute $\lambda = -\frac{\sqrt{6}}{6}$ into l_1. So the centre of the sphere, P, has coordinates $(3, -2, 4)$.

$$\overrightarrow{AP} = \overrightarrow{OP} - \overrightarrow{OA} = \begin{pmatrix} 3 \\ -2 \\ 4 \end{pmatrix} - \begin{pmatrix} 4 \\ -5 \\ 4-\sqrt{6} \end{pmatrix} = \begin{pmatrix} -1 \\ 3 \\ \sqrt{6} \end{pmatrix}.$$

$$|\overrightarrow{AP}| = \sqrt{(-1)^2 + 3^2 + (\sqrt{6})^2} = \sqrt{16} = 4.$$

Volume of sphere $= \frac{4}{3}\pi r^3 = \frac{4}{3}\pi(4)^3 = \frac{256\pi}{3}$ units^3.

42 a)

$$3 + 3\lambda = 5 + 16\mu \Rightarrow 3\lambda - 16\mu = 2 \quad (1), \quad 2 - 3\lambda = -8 - 8\mu \Rightarrow -3\lambda + 8\mu = -10 \quad (2),$$

$$-4 + 5\lambda = 3 + 23\mu \Rightarrow 5\lambda - 23\mu = 7 \quad (3).$$

Solve (1) and (2): $\lambda = 6$ and $\mu = 1$. Check in (3): $5(6) - 23(1) = 7$ as required. So the lines intersect. Substitute $\lambda = 6$ into l_1. The lines intersect at $P(21, -16, 26)$.

42 b)

$$\overrightarrow{AP} = \overrightarrow{OP} - \overrightarrow{OA} = \begin{pmatrix} 21 \\ -16 \\ 26 \end{pmatrix} - \begin{pmatrix} 3+3\lambda \\ 2-3\lambda \\ -4+5\lambda \end{pmatrix} = \begin{pmatrix} 18-3\lambda \\ -18+3\lambda \\ 30-5\lambda \end{pmatrix}.$$

$$|\overrightarrow{AP}| = 849\sqrt{43} \Rightarrow \sqrt{(18-3\lambda)^2 + (-18+3\lambda)^2 + (30-5\lambda)^2} = 849\sqrt{43}$$

$$\Rightarrow 324 - 108\lambda + 9\lambda^2 + 324 - 108\lambda + 9\lambda^2 + 900 - 300\lambda + 25\lambda^2 = 30994443$$

$$\Rightarrow 43\lambda^2 - 516\lambda - 30992895 = 0 \Rightarrow \lambda^2 - 12\lambda - 720765 = 0 \Rightarrow (\lambda + 843)(\lambda - 855) = 0$$

$$\Rightarrow \lambda = -843 \text{ or } \lambda = 855.$$

So A has coordinates $(3 + 3(-843), 2 - 3(-843), -4 + 5(-843)) = (-2526, 2531, -4219)$ or $(3 + 3(855), 2 - 3(855), -4 + 5(855)) = (2568, -2563, 4271)$. If $A(-2526, 2531, -4219)$ then

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} 5+16\mu \\ -8-8\mu \\ 3+23\mu \end{pmatrix} - \begin{pmatrix} -2526 \\ 2531 \\ -4219 \end{pmatrix} = \begin{pmatrix} 2531+16\mu \\ -2539-8\mu \\ 4222+23\mu \end{pmatrix}.$$

$$\begin{pmatrix} 16 \\ -8 \\ 23 \end{pmatrix} \cdot \begin{pmatrix} 2531+16\mu \\ -2539-8\mu \\ 4222+23\mu \end{pmatrix} = 0$$

$$\Rightarrow 16(2531 + 16\mu) + (-8)(-2539 - 8\mu) + 23(4222 + 23\mu) = 0$$

$$\Rightarrow 40496 + 256\mu + 20312 + 64\mu + 97106 + 529\mu = 0 \Rightarrow 849\mu + 157914 = 0 \Rightarrow \mu = -186.$$

So B has coordinates $(5 + 16(-186), -8 - 8(-186), 3 + 23(-186)) = (-2971, 1480, -4275)$. If $A(2568, -2563, 4271)$ then

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} 5+16\mu \\ -8-8\mu \\ 3+23\mu \end{pmatrix} - \begin{pmatrix} 2568 \\ -2563 \\ 4271 \end{pmatrix} = \begin{pmatrix} -2563+16\mu \\ 2555-8\mu \\ -4268+23\mu \end{pmatrix}.$$

$$\begin{pmatrix} 16 \\ -8 \\ 23 \end{pmatrix} \cdot \begin{pmatrix} -2563+16\mu \\ 2555-8\mu \\ -4268+23\mu \end{pmatrix} = 0$$

$$\Rightarrow 16(-2563 + 16\mu) + (-8)(2555 - 8\mu) + 23(-4268 + 23\mu) = 0$$

$$\Rightarrow -41008 + 256\mu - 20440 + 64\mu - 98164 + 529\mu = 0 \Rightarrow 849\mu - 159612 = 0 \Rightarrow \mu = 188.$$

So B has coordinates $(5 + 16(188), -8 - 8(188), 3 + 23(188)) = (3013, -1512, 4327)$. So either $B(-2971, 1480, -4275)$ or $B(3013, -1512, 4327)$.

43 a)

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} -1 \\ 3 \\ -2 \end{pmatrix} - \begin{pmatrix} 3 \\ 4 \\ -5 \end{pmatrix} = \begin{pmatrix} -4 \\ -1 \\ 3 \end{pmatrix}.$$

$$\therefore l : \mathbf{r} = \begin{pmatrix} 3 \\ 4 \\ -5 \end{pmatrix} + \lambda \begin{pmatrix} -4 \\ -1 \\ 3 \end{pmatrix}.$$

$$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA} = \begin{pmatrix} 3 \\ p \\ q \end{pmatrix} - \begin{pmatrix} 3 \\ 4 \\ -5 \end{pmatrix} = \begin{pmatrix} 0 \\ p-4 \\ q+5 \end{pmatrix}.$$

$$|\overrightarrow{AC}| = \sqrt{90} \Rightarrow \sqrt{0^2 + (p-4)^2 + (q+5)^2} = \sqrt{90} \Rightarrow (p-4)^2 + (q+5)^2 = 90 \quad (1).$$

$$\begin{pmatrix} -4 \\ -1 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ p-4 \\ q+5 \end{pmatrix} = 0$$

$$\Rightarrow (-4) \times 0 + (-1) \times (p-4) + 3 \times (q+5) = 0 \Rightarrow 0 - p + 4 + 3q + 15 = 0 \Rightarrow p = 3q + 19 \quad (2).$$

Substitute (2) into (1):

$$(3q+15)^2 + (q+5)^2 = 90 \Rightarrow 9q^2 + 90q + 225 + q^2 + 10q + 25 = 90$$

$$\Rightarrow 10q^2 + 100q + 160 = 0 \Rightarrow q^2 + 10q + 16 = 0 \Rightarrow (q+8)(q+2) = 0 \Rightarrow q = -8 \text{ or } q = -2.$$

If $q = -8$ then $p = 3(-8) + 19 = -5$. If $q = -2$ then $p = 3(-2) + 19 = 13$. As $p > 0$, we must have $C(3, 13, -2)$.

43 b)

$$|\overrightarrow{AB}| = \sqrt{(-4)^2 + (-1)^2 + 3^2} = \sqrt{26}.$$

$$l : \mathbf{r} = \begin{pmatrix} 3 \\ 4 \\ -5 \end{pmatrix} + \lambda \begin{pmatrix} -4 \\ -1 \\ 3 \end{pmatrix}.$$

D has coordinates $(3 - 4\lambda, 4 - \lambda, -5 + 3\lambda)$ for some λ.

$$\overrightarrow{AD} = \overrightarrow{OD} - \overrightarrow{OA} = \begin{pmatrix} 3 - 4\lambda \\ 4 - \lambda \\ -5 + 3\lambda \end{pmatrix} - \begin{pmatrix} 3 \\ 4 \\ -5 \end{pmatrix} = \begin{pmatrix} -4\lambda \\ -\lambda \\ 3\lambda \end{pmatrix}.$$

$$|\overrightarrow{AD}| = \sqrt{(-4\lambda)^2 + (-\lambda)^2 + (3\lambda)^2} = \sqrt{26\lambda^2}.$$

$$\text{Area of triangle } CAB = \frac{1}{2}|\overrightarrow{AC}||\overrightarrow{AB}| = \frac{1}{2}\sqrt{90}\sqrt{26}.$$

$$\text{Area of triangle } CAD = \frac{1}{2}|\overrightarrow{AC}||\overrightarrow{AD}| = \frac{1}{2}\sqrt{90}\sqrt{26\lambda^2}.$$

$$\frac{1}{2}\sqrt{90}\sqrt{26\lambda^2} = 2 \times \frac{1}{2}\sqrt{90}\sqrt{26} \Rightarrow \sqrt{26\lambda^2} = 2\sqrt{26} \Rightarrow 26\lambda^2 = 4 \times 26 \Rightarrow \lambda^2 = 4 \Rightarrow \lambda = \pm 2.$$

So $D = (3 - 4(-2), 4 - (-2), -5 + 3(-2)) = (11, 6, -11)$ or $D = (3 - 4(2), 4 - (2), -5 + 3(2)) = (-5, 2, 1)$. Alternative Method:

$$\overrightarrow{OD} = \overrightarrow{OA} \pm 2\overrightarrow{AB} = \begin{pmatrix} 3 \\ 4 \\ -5 \end{pmatrix} \pm 2 \begin{pmatrix} -4 \\ -1 \\ 3 \end{pmatrix} = \begin{pmatrix} -5 \\ 2 \\ 1 \end{pmatrix} \text{ or } \begin{pmatrix} 11 \\ 6 \\ -11 \end{pmatrix}.$$

So D has coordinates $(-5, 2, 1)$ or $(11, 6, -11)$.

44 a)

$$\left(\frac{-1 + 3\sqrt{89} - 1 - 3\sqrt{89}}{2}, \frac{5 - 2\sqrt{89} + 5 + 2\sqrt{89}}{2}, \frac{9 + 6\sqrt{89} + 9 - 6\sqrt{89}}{2} \right) = (-1, 5, 9).$$

44 b)

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} -1-3\sqrt{89} \\ 5+2\sqrt{89} \\ 9-6\sqrt{89} \end{pmatrix} - \begin{pmatrix} -1+3\sqrt{89} \\ 5-2\sqrt{89} \\ 9+6\sqrt{89} \end{pmatrix} = \begin{pmatrix} -6\sqrt{89} \\ 4\sqrt{89} \\ -12\sqrt{89} \end{pmatrix}.$$

$$l_1 : \mathbf{r} = \begin{pmatrix} -1 \\ 5 \\ 9 \end{pmatrix} + \lambda \begin{pmatrix} -6\sqrt{89} \\ 4\sqrt{89} \\ -12\sqrt{89} \end{pmatrix}.$$

To make the numbers nicer, we'll choose to use $l_1 : \mathbf{r} = \begin{pmatrix} -1 \\ 5 \\ 9 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ -2 \\ 6 \end{pmatrix}$.

$$-1+3\lambda = 3+2\mu \Rightarrow 3\lambda - 2\mu = 4 \quad (1), \quad 5-2\lambda = 23+9\mu \Rightarrow -2\lambda - 9\mu = 18 \quad (2),$$

$$9+6\lambda = 13+2\mu \Rightarrow 6\lambda - 2\mu = 4 \quad (3).$$

Solve (1) and (2): $\lambda = 0$ and $\mu = -2$. Check in (3): $6(0) - 2(-2) = 4$ as required. So the lines intersect. Substitute $\lambda = 0$ into l_1. The lines intersect at $(-1, 5, 9)$.

44 c)

$$|\overrightarrow{AP}| = \frac{1}{2}|\overrightarrow{AB}| = \frac{1}{2}\sqrt{(-6\sqrt{89})^2 + (4\sqrt{89})^2 + (-12\sqrt{89})^2}$$

$$= \frac{1}{2}\sqrt{3204 + 1424 + 12816} = \frac{1}{2}\sqrt{17444} = 7\sqrt{89}.$$

$$\overrightarrow{PC} = \overrightarrow{OC} - \overrightarrow{OP} = \begin{pmatrix} 3+2\mu \\ 23+9\mu \\ 13+2\mu \end{pmatrix} - \begin{pmatrix} -1 \\ 5 \\ 9 \end{pmatrix} = \begin{pmatrix} 4+2\mu \\ 18+9\mu \\ 4+2\mu \end{pmatrix}.$$

$$|\overrightarrow{PC}| = 7\sqrt{89} \Rightarrow \sqrt{(4+2\mu)^2 + (18+9\mu)^2 + (4+2\mu)^2} = 7\sqrt{89}$$

$$\Rightarrow 16 + 16\mu + 4\mu^2 + 324 + 324\mu + 81\mu^2 + 16 + 16\mu + 4\mu^2 = 4361 \Rightarrow 89\mu^2 + 356\mu - 4005 = 0$$

$$\Rightarrow \mu^2 + 4\mu - 45 = 0 \Rightarrow (\mu+9)(\mu-5) = 0 \Rightarrow \mu = -9 \text{ or } \mu = 5.$$

So C and D have coordinates $(3+2(-9), 23+9(-9), 13+2(-9)) = (-15, -58, -5)$ or $(3+2(5), 23+9(5), 13+2(5)) = (13, 68, 23)$.

45)

$P(18, 4, -1)$ lies on l_3, and l_3 is parallel to l_1, so

$$l_3 : \mathbf{r} = \begin{pmatrix} 18 \\ 4 \\ -1 \end{pmatrix} + \alpha \begin{pmatrix} 4 \\ -2 \\ 1 \end{pmatrix}.$$

Q lies on l_3 and so has coordinates $(18+4\alpha, 4-2\alpha, -1+\alpha)$ for some α. R lies on l_2 when $\mu = -27$ and so has coordinates

$$(0 + (-27) \times (-6), 7 + (-27) \times 1, 5 + (-27) \times 2) = (162, -20, -49).$$

$$\overrightarrow{RQ} = \overrightarrow{OQ} - \overrightarrow{OR} = \begin{pmatrix} 18+4\alpha \\ 4-2\alpha \\ -1+\alpha \end{pmatrix} - \begin{pmatrix} 162 \\ -20 \\ -49 \end{pmatrix} = \begin{pmatrix} -144+4\alpha \\ 24-2\alpha \\ 48+\alpha \end{pmatrix}.$$

$\overrightarrow{RQ}$ is perpendicular to l_2, so

$$\begin{pmatrix}-6\\1\\2\end{pmatrix}\cdot\begin{pmatrix}-144+4\alpha\\24-2\alpha\\48+\alpha\end{pmatrix}=0$$

$$\Rightarrow(-6)(-144+4\alpha)+1(24-2\alpha)+2(48+\alpha)=0$$
$$\Rightarrow 864-24\alpha+24-2\alpha+96+2\alpha=0$$
$$\Rightarrow 984-24\alpha=0$$
$$\Rightarrow\alpha=41.$$
$$\therefore Q=(18+4(41),4-2(41),-1+(41))=(182,-78,40).$$

46 a)

Drone 1 travels from $(0,12,1)$ to $(15,6,15)$.

$$\begin{pmatrix}15\\6\\15\end{pmatrix}-\begin{pmatrix}0\\12\\1\end{pmatrix}=\begin{pmatrix}15\\-6\\14\end{pmatrix},\quad l_1:\mathbf{r}=\begin{pmatrix}0\\12\\1\end{pmatrix}+\lambda\begin{pmatrix}15\\-6\\14\end{pmatrix}.$$

Drone 2 travels from $(0,3,4)$ to $(15,9,k)$.

$$\begin{pmatrix}15\\9\\k\end{pmatrix}-\begin{pmatrix}0\\3\\4\end{pmatrix}=\begin{pmatrix}15\\6\\k-4\end{pmatrix},\quad l_2:\mathbf{r}=\begin{pmatrix}0\\3\\4\end{pmatrix}+\mu\begin{pmatrix}15\\6\\k-4\end{pmatrix}.$$

$$0+15\lambda=0+15\mu\Rightarrow\lambda=\mu\quad(1),\quad 12-6\lambda=3+6\mu\Rightarrow-6\lambda-6\mu=-9\quad(2),$$
$$1+14\lambda=4+\mu(k-4)\quad(3).$$

Solve (1) and (2): $\lambda=\frac{3}{4}$ and $\mu=\frac{3}{4}$. Substitute into (3):

$$1+14(\frac{3}{4})=4+(\frac{3}{4})(k-4)\Rightarrow 1+\frac{21}{2}=4+\frac{3}{4}k-3\Rightarrow k=14.$$

46 b)

Distance from $(0,12,1)$ to $(15,6,15)$ is

$$\sqrt{(15-0)^2+(6-12)^2+(15-1)^2}=\sqrt{457}\text{ metres.}$$

So Drone 1 travels $\sqrt{457}$ metres in 5 seconds, so it is travelling at a constant speed of $\frac{1}{5}\sqrt{457}$ ms^{-1}.
Point of intersection of l_1 and l_2:

$$\left(0+(\frac{3}{4})\times 15,12+(\frac{3}{4})\times(-6),1+(\frac{3}{4})\times 14\right)=\left(\frac{45}{4},\frac{15}{2},\frac{23}{2}\right).$$

Distance from $(0,12,1)$ to $(\frac{45}{4},\frac{15}{2},\frac{23}{2})$ is

$$\sqrt{(\frac{45}{4}-0)^2+(\frac{15}{2}-12)^2+(\frac{23}{2}-1)^2}=\sqrt{\frac{4113}{4}}=\frac{3}{4}\sqrt{457}\text{ metres.}$$

Drone 1 travels from its initial starting position to the point of intersection in $\frac{\frac{3}{4}\sqrt{457}}{\frac{1}{5}\sqrt{457}}=\frac{15}{4}$ seconds.
Distance from $(0,3,4)$ to $(\frac{45}{4},\frac{15}{2},\frac{23}{2})$ is

$$\sqrt{(\frac{45}{4}-0)^2+(\frac{15}{2}-3)^2+(\frac{23}{2}-4)^2}=\sqrt{\frac{3249}{16}}=\frac{57}{4}\text{ metres.}$$

So Drone 2 must travel $\frac{57}{4}$ metres in $\frac{15}{4}$ seconds to crash into Drone 1, so its speed must be $\frac{57/4}{15/4}=\frac{19}{5}=3.8$ ms^{-1}.

46 c)

The drones are modelled as particles, with negligible size, so they could crash into one another sooner. We have also assumed that the drones instantaneously reach their constant speed.

47 a)

$$\begin{pmatrix}9\\3\\-3\end{pmatrix}-\begin{pmatrix}6\\2\\-2\end{pmatrix}=\begin{pmatrix}3\\1\\-1\end{pmatrix},\quad \begin{pmatrix}6\\2\\-2\end{pmatrix}-2\begin{pmatrix}3\\1\\-1\end{pmatrix}=\begin{pmatrix}0\\0\\0\end{pmatrix}.$$

So the origin lies on the line through A and B. So $\mathbf{r}=\begin{pmatrix}0\\0\\0\end{pmatrix}+\lambda\begin{pmatrix}3\\1\\-1\end{pmatrix}=\lambda\begin{pmatrix}3\\1\\-1\end{pmatrix}$ as required.

47 b)

$$\overrightarrow{OC}=\begin{pmatrix}0&-1&0\\1&0&0\\0&0&1\end{pmatrix}\begin{pmatrix}6\\2\\-2\end{pmatrix}=\begin{pmatrix}-2\\6\\-2\end{pmatrix},\quad \overrightarrow{OD}=\begin{pmatrix}0&-1&0\\1&0&0\\0&0&1\end{pmatrix}\begin{pmatrix}9\\3\\-3\end{pmatrix}=\begin{pmatrix}-3\\9\\-3\end{pmatrix}.$$

$$\overrightarrow{CD}=\overrightarrow{OD}-\overrightarrow{OC}=\begin{pmatrix}-3\\9\\-3\end{pmatrix}-\begin{pmatrix}-2\\6\\-2\end{pmatrix}=\begin{pmatrix}-1\\3\\-1\end{pmatrix}.$$

$$l:\mathbf{r}=\begin{pmatrix}-2\\6\\-2\end{pmatrix}+\mu\begin{pmatrix}-1\\3\\-1\end{pmatrix}=\begin{pmatrix}0\\0\\0\end{pmatrix}+\mu\begin{pmatrix}-1\\3\\-1\end{pmatrix}=\mu\begin{pmatrix}-1\\3\\-1\end{pmatrix}.$$

47 c)

$$\cos\theta=\frac{\begin{pmatrix}3\\1\\-1\end{pmatrix}\cdot\begin{pmatrix}-1\\3\\-1\end{pmatrix}}{|\begin{pmatrix}3\\1\\-1\end{pmatrix}||\begin{pmatrix}-1\\3\\-1\end{pmatrix}|}=\frac{3\times(-1)+1\times3+(-1)\times(-1)}{\sqrt{3^2+1^2+(-1)^2}\sqrt{(-1)^2+3^2+(-1)^2}}=\frac{1}{\sqrt{11}\sqrt{11}}=\frac{1}{11}.$$

$$\theta=\arccos(\frac{1}{11})=84.8^\circ\ (1\text{ DP}).$$

48)

$$\begin{pmatrix}1&2&0\\-1&5&0\\0&0&3\end{pmatrix}\begin{pmatrix}1+2\lambda\\1-\lambda\\2+3\lambda\end{pmatrix}=\begin{pmatrix}1(1+2\lambda)+2(1-\lambda)+0(2+3\lambda)\\-(1+2\lambda)+5(1-\lambda)+0(2+3\lambda)\\0(1+2\lambda)+0(1-\lambda)+3(2+3\lambda)\end{pmatrix}$$

$$=\begin{pmatrix}3\\4-7\lambda\\6+9\lambda\end{pmatrix}=\begin{pmatrix}3\\4\\6\end{pmatrix}+\lambda\begin{pmatrix}0\\-7\\9\end{pmatrix}.$$

So $l_2:\mathbf{r}=\begin{pmatrix}3\\4\\6\end{pmatrix}+\lambda\begin{pmatrix}0\\-7\\9\end{pmatrix}$.

49)

$$\begin{pmatrix} 2 & 1 & 0 \\ 1 & 3 & 1 \\ 1 & 0 & 2 \end{pmatrix} \begin{pmatrix} 2+3\lambda \\ 3-\lambda \\ -4-\lambda \end{pmatrix} = \begin{pmatrix} 2(2+3\lambda)+1(3-\lambda)+0(-4-\lambda) \\ 1(2+3\lambda)+3(3-\lambda)+1(-4-\lambda) \\ 1(2+3\lambda)+0(3-\lambda)+2(-4-\lambda) \end{pmatrix}$$

$$= \begin{pmatrix} 7+5\lambda \\ 7-\lambda \\ -6+\lambda \end{pmatrix} = \begin{pmatrix} 7 \\ 7 \\ -6 \end{pmatrix} + \lambda \begin{pmatrix} 5 \\ -1 \\ 1 \end{pmatrix}.$$

So $l_2 : \mathbf{r} = \begin{pmatrix} 7 \\ 7 \\ -6 \end{pmatrix} + \lambda \begin{pmatrix} 5 \\ -1 \\ 1 \end{pmatrix}$.

50)

$$\begin{pmatrix} 1 & -2 & 1 \\ 1.5 & -1 & 0.25 \\ 1 & 0.5 & -3 \end{pmatrix} \begin{pmatrix} 5+\lambda \\ 4-\lambda \\ 2+2\lambda \end{pmatrix} = \begin{pmatrix} 1(5+\lambda)-2(4-\lambda)+1(2+2\lambda) \\ 1.5(5+\lambda)-(4-\lambda)+0.25(2+2\lambda) \\ 1(5+\lambda)+0.5(4-\lambda)-3(2+2\lambda) \end{pmatrix}$$

$$= \begin{pmatrix} 5+\lambda-8+2\lambda+2+2\lambda \\ \frac{15}{2}+\frac{3}{2}\lambda-4+\lambda+\frac{1}{2}+\frac{1}{2}\lambda \\ 5+\lambda+2-\frac{1}{2}\lambda-6-6\lambda \end{pmatrix} = \begin{pmatrix} 5\lambda-1 \\ 4+3\lambda \\ 1-\frac{11}{2}\lambda \end{pmatrix} = \begin{pmatrix} -1 \\ 4 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 5 \\ 3 \\ -\frac{11}{2} \end{pmatrix}$$

$$= \begin{pmatrix} -1 \\ 4 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} 10 \\ 6 \\ -11 \end{pmatrix}, \qquad \text{where } \mu = \frac{1}{2}\lambda.$$

51)

$$\begin{pmatrix} p & 1 & -3.5 \\ 3 & -3 & 2 \\ 2.5 & -1 & q \end{pmatrix} \begin{pmatrix} 2+8\lambda \\ 3+2\lambda \\ -6-5\lambda \end{pmatrix} = \begin{pmatrix} 26+a\mu \\ -15+16\mu \\ -10+b\mu \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} p(2+8\lambda)+1(3+2\lambda)-\frac{7}{2}(-6-5\lambda) \\ 3(2+8\lambda)-3(3+2\lambda)+2(-6-5\lambda) \\ \frac{5}{2}(2+8\lambda)-(3+2\lambda)+q(-6-5\lambda) \end{pmatrix} = \begin{pmatrix} 26+a\mu \\ -15+16\mu \\ -10+b\mu \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 2p+8p\lambda+3+2\lambda+21+\frac{35}{2}\lambda \\ 6+24\lambda-9-6\lambda-12-10\lambda \\ 5+20\lambda-3-2\lambda-6q-5q\lambda \end{pmatrix} = \begin{pmatrix} 26+a\mu \\ -15+16\mu \\ -10+b\mu \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} (2p+24)+(8p+\frac{39}{2})\lambda \\ -15+8\lambda \\ (2-6q)+(18-5q)\lambda \end{pmatrix} = \begin{pmatrix} 26+a\mu \\ -15+16\mu \\ -10+b\mu \end{pmatrix}.$$

$$(2p+24)+(8p+\frac{39}{2})\lambda = 26+a\mu \quad (1), \quad -15+8\lambda = -15+16\mu \Rightarrow \lambda = 2\mu \quad (2),$$

$$(2-6q)+(18-5q)\lambda = -10+b\mu \quad (3).$$

Substitute (2) into (1):

$$(2p+24)+(16p+39)\mu = 26+a\mu, \quad 2p+24 = 26 \Rightarrow p = 1, \quad 16\times 1+39 = a \Rightarrow a = 55.$$

Substitute (2) into (3):

$$(2-6q)+(36-10q)\mu = -10+b\mu, \quad 2-6q = -10 \Rightarrow q = 2, \quad 36-10\times 2 = b \Rightarrow b = 16.$$

So $a = 55, b = 16, p = 1, q = 2$.

52)

Let C be the centre of the circle, so $\overrightarrow{OC} = \begin{pmatrix}1\\1\\0\end{pmatrix}$. Let P be a general point on the line l, so $\overrightarrow{OP} = \begin{pmatrix}-2+4\lambda\\3+2\lambda\\1+\lambda\end{pmatrix}$.

$$\overrightarrow{CP} = \overrightarrow{OP} - \overrightarrow{OC} = \begin{pmatrix}-2+4\lambda\\3+2\lambda\\1+\lambda\end{pmatrix} - \begin{pmatrix}1\\1\\0\end{pmatrix} = \begin{pmatrix}-3+4\lambda\\2+2\lambda\\1+\lambda\end{pmatrix}.$$

$$|\overrightarrow{CP}| = \sqrt{14} \Rightarrow (-3+4\lambda)^2 + (2+2\lambda)^2 + (1+\lambda)^2 = 14$$

$$\Rightarrow 9 - 24\lambda + 16\lambda^2 + 4 + 8\lambda + 4\lambda^2 + 1 + 2\lambda + \lambda^2 = 14 \Rightarrow 21\lambda^2 - 14\lambda = 0$$

$$\Rightarrow 3\lambda^2 - 2\lambda = 0 \Rightarrow (3\lambda - 2)\lambda = 0 \Rightarrow \lambda = 0 \text{ or } \lambda = \frac{2}{3}.$$

When $\lambda = 0$, $\overrightarrow{OP} = (-2, 3, 1)$. When $\lambda = \frac{2}{3}$, $\overrightarrow{OP} = (\frac{2}{3}, \frac{13}{3}, \frac{5}{3})$. So the circle intersects l at $(-2, 3, 1)$ and $(\frac{2}{3}, \frac{13}{3}, \frac{5}{3})$.

53)

Show that $(40, 7, -20)$ lies in the plane:

$$3(40) - 2(7) + 5(-20) = 120 - 14 - 100 = 6 \text{ as required.}$$

Show that $\begin{pmatrix}-4\\9\\6\end{pmatrix}$ is perpendicular to the plane:

$$\begin{pmatrix}-4\\9\\6\end{pmatrix} \cdot \begin{pmatrix}3\\-2\\5\end{pmatrix} = (-4) \times 3 + 9 \times (-2) + 6 \times 5 = -12 - 18 + 30 = 0.$$

So the line l lies in the plane Π.

54)

$$\Pi : \mathbf{r} = \begin{pmatrix}9\\-1\\3\end{pmatrix} + \lambda \begin{pmatrix}3\\-2\\5\end{pmatrix} + \mu \begin{pmatrix}9\\-3\\-8\end{pmatrix}.$$

55)

$$\begin{pmatrix}4\\6\\-3\end{pmatrix} \times \begin{pmatrix}2\\9\\18\end{pmatrix} = \begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\4 & 6 & -3\\2 & 9 & 18\end{vmatrix}$$

$$= (6 \times 18 - (-3) \times 9)\mathbf{i} - (4 \times 18 - (-3) \times 2)\mathbf{j} + (4 \times 9 - 6 \times 2)\mathbf{k} = 135\mathbf{i} - 78\mathbf{j} + 24\mathbf{k}.$$

So the equation of the plane Π is $135x - 78y + 24z = d$. Substitute in $(1, -6, 7)$ to find d:

$$135(1) - 78(-6) + 24(7) = d \Rightarrow d = 771.$$

$$\therefore 135x - 78y + 24z = 771.$$

56)

$$\begin{pmatrix}2\\1\\9\end{pmatrix}\times\begin{pmatrix}-3\\5\\6\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\2&1&9\\-3&5&6\end{vmatrix}$$

$$=(1\times 6-9\times 5)\mathbf{i}-(2\times 6-9\times(-3))\mathbf{j}+(2\times 5-1\times(-3))\mathbf{k}=-39\mathbf{i}-39\mathbf{j}+13\mathbf{k}.$$

So the equation of the plane Π is $-39x-39y+13z=d$. Substitute in $(2,6,-1)$ to find d:

$$-39(2)-39(6)+13(-1)=d\Rightarrow d=-325.$$

$$\therefore -39x-39y+13z=-325.$$

Substitute in $(4,k,-5)$ to find k:

$$-39(4)-39k+13(-5)=-325\Rightarrow -156-39k-65=-325\Rightarrow 104-39k=0\Rightarrow k=\frac{8}{3}.$$

57)

$$\begin{pmatrix}k\\3\\1\end{pmatrix}\times\begin{pmatrix}-1\\-2\\8\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\k&3&1\\-1&-2&8\end{vmatrix}$$

$=(3\times 8-1\times(-2))\mathbf{i}-(k\times 8-1\times(-1))\mathbf{j}+(k\times(-2)-3\times(-1))\mathbf{k}=26\mathbf{i}+(-8k-1)\mathbf{j}+(-2k+3)\mathbf{k}.$

So the equation of the plane Π is $26x+(-8k-1)y+(-2k+3)z=d$. Substitute in $(2,3,-8)$:

$$26(2)+(-8k-1)(3)+(-2k+3)(-8)=d\Rightarrow 52-24k-3+16k-24=d\Rightarrow 25-8k=d\quad(1).$$

Substitute in $(4,1,10)$:

$$26(4)+(-8k-1)(1)+(-2k+3)(10)=d\Rightarrow 104-8k-1-20k+30=d\Rightarrow 133-28k=d\quad(2).$$

Solve (1) and (2): $k=\frac{27}{5}$ and $d=-\frac{91}{5}$. So the equation of the plane is

$$26x+(-8(\frac{27}{5})-1)y+(-2(\frac{27}{5})+3)z=-\frac{91}{5}\Rightarrow 26x-\frac{221}{5}y-\frac{39}{5}z=-\frac{91}{5}\Rightarrow 130x-221y-39z=-91.$$

58 a)

$$\frac{x-0}{2}=\frac{y-0}{5}=\frac{z-0}{-7}$$

or $\frac{x}{2}=\frac{y}{5}=\frac{z}{-7}$.

58 b)

$$\frac{x+2}{4}=\frac{y-3}{-1}=\frac{z-5}{1}$$

or $\frac{x+2}{4}=3-y=z-5$.

58 c)

$$x = 4, \qquad \frac{y+8}{1} = \frac{z-0}{6}$$

or $x = 4, \quad y + 8 = \frac{z}{6}$.

58 d)

$$\frac{x-0}{3} = \frac{z-7}{-8}, \qquad y = 1$$

or $\frac{x}{3} = \frac{7-z}{8}, \quad y = 1$.

58 e)

$$\frac{x+9}{1} = \frac{y-3}{-5}, \qquad z = 8$$

or $x + 9 = \frac{3-y}{5}, \quad z = 8$.

58 f)

$x + 2 = \lambda, \quad y = 1, \quad z = -6$.

59 a)

$$(\mathbf{r} - \begin{pmatrix} 2 \\ -3 \\ 4 \end{pmatrix}) \times \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix} = \mathbf{0}.$$

59 b)

$$(\mathbf{r} - \begin{pmatrix} 0 \\ -6 \\ 5 \end{pmatrix}) \times \begin{pmatrix} 4 \\ -2 \\ -5 \end{pmatrix} = \mathbf{0}.$$